

LAVAN REDDY MANDADI

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PROFESSIONAL SUMMARY

Results-driven Automotive Quality and Process Engineer with over 5 years of experience in automotive manufacturing, quality assurance, and process improvement. Proficient in IATF 16949, APQP, PPAP, FMEA, MSA, SPC, and Control Plan development to ensure compliance and production readiness. Skilled in process validation, Cpk analysis, root cause analysis, CAPA, and continuous improvement using Lean Six Sigma methodologies. Experienced in CNC machining, assembly operations, supplier quality management, and new product introduction (NPI). Proven success in improving First Pass Yield, reducing defects, and enhancing overall equipment effectiveness (OEE) through data-driven problem-solving and cross-functional collaboration.

TECHNICAL SKILLS

- **Quality & Compliance:** ISO 13485, ISO 9001, FDA 21 CFR Part 11, EU-MDR, CDSCO, QSR
- **Risk & Audits:** ISO 14971, Design FMEA, Process FMEA, Risk Management, CAPA, Control Plans
- **Manufacturing Quality:** APQP, PPAP, MSA, SPC, DOE, Gage R&R
- **Continuous Improvement:** 8D Problem Solving, Root Cause Analysis, Lean, Kaizen, 5S, Value Stream Mapping
- **Process Validation:** IQ, OQ, PQ, Protocols, COQ Analysis
- **Regulatory Support:** FDA Audit Readiness, Supplier CAR, Vendor Qualification
- **Manufacturing Processes:** CNC Machining, Injection Molding, Laser Cutting, Assembly, ETO Sterilization
- **Production Support:** Fixture Design, Cleanroom Operations, Poka-Yoke Implementation
- **Tools & Platforms:** SolidWorks, AutoCAD, Minitab, Power BI, SAP, MS Office Suite
- **Systems & Software:** QMS Software, CMM Inspection, eBR Systems, Excel Dashboards
- **Certifications:** Lean Six Sigma Green Belt (2024), ISO 13485 QMS, FDA QSR & 21 CFR Part 11
- **Specialized Training:** Design Controls, FMEA, First Article Inspection (FAI), AS9102 Standards

WORK EXPERIENCE

Manufacturing Quality Engineer, Etezazi Industries, Aug 2025 – present

- Proficient in G-code and M-code programming for 3-axis and 5-axis CNC machines, ensuring precise tool paths and optimized cycle times.
- Experienced in resolving machine, toolpath, and program-related issues to minimize downtime and maintain high productivity.
- Hands-on knowledge of cutting tools such as end mills, ball nose cutters, reamers, and drills, optimizing tool selection for material and geometry requirements.
- Skilled in using Zoller tool presetters for accurate tool offset and measurement, reducing setup time and improving repeatability.
- Perform OEE (Overall Equipment Effectiveness) and First Pass Yield analysis to monitor machine performance and identify areas for improvement.
- Collaborate with production and quality teams to implement corrective and preventive actions (CAPA) for machining defects.
- Plan and balance machine capacity and workload, improving utilization and throughput.
- Support process improvement initiatives using Lean and Six Sigma tools to enhance efficiency and reduce scrap.
- Maintain and update CNC programs to improve tool life, part quality, and machining efficiency.
- Conduct root cause analysis (using tools like 5 Whys and Fishbone diagrams) for machining-related quality issues

Quality & Continuous Improvement Engineer

Etezazi Industries, Wichita, KS, USA

June 2024 – Aug 2025

- **Manage Customer Quality Requirements:** Identify, interpret, and communicate customer contractual requirements to ensure compliance across all manufacturing processes.
- **Customer Interface & Support:** Serve as the primary interface for quality-related activities, including handling OEM customer complaints from Boeing, Northrop Grumman, Spirit, and other clients.

- **Audits & Compliance:** Host and support customer and industry quality audits and certifications, while conducting internal process and product audits to maintain compliance with established standards.
- **Inspection & First Article Planning:** Develop and maintain inspection plans and methods, including First Article Inspection (FAI) plans, and ensure appropriate inspection tools and equipment are used efficiently.
- **Non-Conformance & Corrective Actions:** Evaluate, document, and disposition non-conforming products through MRB activities, and drive 8D problem-solving to analyze root causes and implement preventive actions.
- **Process & Product Design Support:** Participate in design reviews, design verification, and Integrated Product Team (IPT) activities; perform FMEA and APQP at early stages to ensure robust quality planning.
- **Training & Continuous Improvement:** Guide and train operators on quality procedures, inspection criteria, and process improvements; lead team-based corrective actions to enhance process capability and performance

Process Quality Engineer

Tata Motors, Hyderabad, TG, India Oct 2020 – Dec 2022

- Developed and optimized manufacturing processes for precision automotive components, achieving process capability ($Cpk \geq 1.33$).
- Led the APQP roadmap for new product introduction from concept to production validation and run-at-rate.
- Conducted MSA and Gage R&R studies, reducing measurement variation from 28% to 12%.
- Validated key machining and assembly processes in alignment with IATF 16949 and customer-specific requirements.
- Authored and maintained 18 work instructions and standard operating procedures (SOPs) with visual aids to enhance operator training and compliance.
- Facilitated Process FMEAs (PFMEA) and implemented risk mitigation and control measures across production lines.
- Audited and qualified 7 new suppliers based on PPAP and process capability requirements.
- Implemented poka-yoke and error-proofing solutions, reducing in-process assembly defects by 72%.
- Led multiple 5S and lean improvement projects on the shop floor, improving traceability and process efficiency.
- Applied DMAIC methodology to resolve scheduling bottlenecks, identified root causes, and implemented effective corrective and preventive actions.

Junior Manufacturing Quality Engineer

Azad Engineering Pvt Ltd., Hyderabad, TG, India

Jun 2019 - Sep 2020

- Supported CNC process setup for turbine and orthopedic parts (94% FPY).
- Performed incoming inspection for 600+ precision components monthly.
- Participated in Design FMEA and recommended controls for bore defects.
- Reduced machining defects by 55% through spindle and tool optimization.
- Created first-piece inspection and CMM GR&R procedures.
- Verified heat-treatment and passivation processes for AMS compliance.
- Implemented Lean 5S practices across manufacturing floor.
- Assisted senior manufacturing quality engineer in recording data and analyzing OEE

EDUCATION

- **Master of Science, Industrial Engineering** | Wichita State University, Wichita, KS, USA
- **Bachelor of Technology, Mechanical Engineering** | TKR College of Engineering & Technology, Hyderabad, TG, India